

Fabrizio Sossan – Curriculum Vitae

Full Professor of Electrical Power Systems and Head of Gridlab
 Haute Ecole Spécialisée de Suisse Occidentale (HES-SO Valais-Wallis)
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Links: [Personal webpage](#) | [Google Scholar](#) | [Research Gate](#) | [Scopus](#) | [ORCID iD](#)

Research Interests and Activities

Research interests include power systems, energy storage, hydropower flexibility, and integrated energy systems. My research focuses on the modeling, optimization, and control of energy systems, with particular emphasis on flexibility provision and the integration of renewable energy resources. I lead Gridlab, the power systems laboratory of HES-SO Valais-Wallis, and a research team averaging 10 junior and senior researchers, engineers, and doctoral students. To date, my research activities have attracted more than CHF 6 million in competitive external funding from European, national, and industrial sources. I maintain active collaborations with utilities, grid operators, technology providers, and industrial partners across the electricity value chain, both nationally and internationally.

Education

08/2014	Ph.D. in Electrical Engineering, Technical University of Denmark (DTU), Denmark, with the thesis “Indirect control of flexible demand for power system applications”.
02/2010	M.Sc. in Computer Engineering (mark: 110/110), University of Genova, Italy.

Academic Appointments and Professional Experience

09/2025 – present	HES-SO VALAIS-WALLIS, Full Professor of Electrical Power Systems (Switzerland)
11/2022 – 08/2025	HES-SO VALAIS-WALLIS, Associate Professor of Power Systems (Switzerland)
03/2019 – 09/2022	MINES PARIS-PSL, Associate Professor of Renewable Energies (France)
08/2018 – 02/2019	ETHZ, Researcher (Switzerland)
11/2017 – 07/2018	NREL, Guest researcher (USA)
05/2014 – 10/2017	EPFL, Researcher (Switzerland)
01/2011 – 04/2014	DTU, PhD student (Denmark)
10/2010 – 12/2010	RISØ, research assistant (Denmark)
03/2010 – 09/2010	ANSALDO SISTEMI INDUSTRIALI, Automation Engineer (Italy)
01/2008 – 01/2010	SELF-EMPLOYED DURING STUDIES, Programmer (Italy)

Leadership and Professional Appointments

2025 – present	Head of Gridlab, HES-SO Valais-Wallis
2026 – present	Member, Swiss National Committee of CIGRE

Compiled on June 3, 2026.

2025 – present	Member, SFOE Trendwatching Group "Grids and Flexibility (NetFlex)"
2025 – present	Advisory Board Member, Inova Energy
2021 – 2022	Director, Specialized Master "Alternative Energies for the Future", MINES Paris
2020 – 2022	Co-founder and ad interim CTO, ModBESS

Research Funding

Total research funding secured as Project Owner, PI, or Co-PI: > CHF 6 million. Average annual external competitive funding: approx. CHF 850,000.

2024 – 2027	(PI and WP leader) Innovative Storage Technology And Operations In Hydropower, (STOR-HY), Horizon Europe project. Fund: CHF 1.0 million.
2023 – 2027	(Project owner and PI) Energy Storage Infrastructure for 100% Production from Renewables and Energy Self-Sufficiency in Switzerland (STORE), Innosuisse Flagship 108.230. Fund: CHF 4.2 million.
2023 – 2024	(PI) More resilient microgrids and grids (ResiNet). Fund: CHF 220'000 (my share: CHF 70'000).
2022	(Project owner and PI) Financing of Laboratory on Integrated Energy Systems, Region Sud - Provence Alpes Cote d'Azur. Fund: € 300'000.
2019 – 2023	(PI) Hydropower Extending Power System Flexibility (XFLEX Hydro), H2020 innovation action. Fund: € 285'000.
2019 – 2021	(PI and WP leader) Optimization of Regional Infrastructures for the Transition to Electric and Connected Autonomous Vehicles (EVA), ERA-NET SES project. Fund: € 100'000.
2018 – 2021	(Main author) Licensing of patent application 62/354,828 to Eaton corporation. Fund: € 30'000 per year.

Supervised Ph.D. students

2024 – present	1. Ali Mohammed Ali (HES-SO Valais-Wallis, Geneva University). Shared supervision with Prof. M. Patel.
2024 – present	2. Keju Jia (HES-SO Valais-Wallis, EPFL). Shared supervision with Prof. M. Paolone.
2020 – 2023	3. Biswarup Mukherjee (MINES ParisTech), Optimization methods for scheduling the charge of electric vehicles and planning their charging infrastructure. Shared supervision with G. Kariniotakis. ¹
2019 – 2023	4. Dr. Stefano Cassano (MINES ParisTech), Control and scheduling of hybrid hydropower plants with batteries for enhanced flexibility in future power systems. Shared supervision with G. Kariniotakis. ¹

Co-supervised Ph.D. students

2019 – 2022	1. Dr. Rahul Gupta (EPFL), Dispatching controllable resources in low-voltage power grids (supervisor: Prof. M. Paolone).
2017 – 2021	2. Dr. Yihui Zuo (EPFL), Impact of Battery Energy Storage Systems on the Dynamic Behavior of Low-inertia Power Grids (supervisor: Prof. M. Paolone).

¹Due to the regulations of the doctoral school, my name does not appear on the cover page of the publicly available thesis manuscript as a supervisor. A formal certificate attesting my supervisory role is available at this [link](#).

Teaching

2026 – present	Fundamentals of Mathematical Optimization for Electrical Systems (2 ECTS), HES-SO Valais-Wallis.
2025 – present	Power Systems Analysis (3 ECTS), HES-SO MSE.
2024 – present	Energy Storage Systems (3 ECTS, EPFL).
2022 – present	Électricité de base (5 ECTS), Réseaux Électriques 1 (2 ECTS), and Réseaux Électriques 2 (6 ECTS), HES-SO Valais-Wallis.
2021 – 2022	Introduction to Power Systems (16 h/semester), MINES ParisTech / PSL.
2021	Energy Storage for Renewable-Based Power Systems (guest lecture), MINES ParisTech.
2019 – 2021	AC Circuits and Phasors and Renewable Energy Technologies , MINES ParisTech.

Invited presentations and seminars

2025	19. Sossan, F. (2025 November). Photovoltaïque à grande échelle : quels défis pour le réseau ? Swissgrid seminar on integration of renewables, Switzerland (Sion).
2025	18. Sossan, F. (2025 October). Organizer and convenor of the event Harnessing Hydropower's Energy Storage and Flexibility for the Energy Transition, Switzerland (Sion).
2025	17. Sossan, F. (2025 April). Le rôle du stockage de l'énergie dans les réseaux électriques. Carrefour de la transition, Switzerland (Sion).
2024	16. Sossan, F. (2024 November). La flexibilité comme alternative au renforcement du réseau (in French). Workshop on Flexibilité des Réseaux de Distribution, Switzerland (Yverdon-les-Bains).
2024	15. Sossan, F. (2024 October). Infrastruttura di stoccaggio di energia per una produzione 100% da rinnovabili in Svizzera. Seminar, VSE/AES Multidis meeting with the directors of Swiss-Italian DSOs, Switzerland (Lugano).
2024	14. Sossan, F. , & Roduit, P. (2024 September). Flexibilité dans les réseaux de distribution. Seminar, Multidis meeting with the directors of Swiss-French DSOs, Switzerland.
2024	13. Sossan, F. (2024 September). Energy Storage Infrastructure for 100% renewable generation. Seminar, VSE/AES meeting with the directors of Swiss-French DSOs, Switzerland (Martigny).
2024	12. Sossan, F. (2024 September). Experimental validation of grid-forming converters in a low-inertia setting. Invited talk at DynPower 2024, Switzerland (Aarau).
2024	11. Sossan, F. (2024 July). Increasing energy storage capacity of hydropower plants: a perspective on quick ramping rates. Invited talk at the workshop "The role of storage capacity of hydropower plants" organized by the PEN@Hydropower association. Online.
2022	10. Sossan, F. (2022 March). Stress-informed model predictive control of hybrid hydropower. Seminar, Waterloo Institute Sustainable Energy (WISE) and University of Waterloo, Canada.
2020	9. Sossan, F. (2020 August). Leveraging autonomous driving of electric vehicles to provide ancillary services to the distribution grid. Panel presentation, session "Electric Vehicles as Flexible Demand-side Resources: Research Progress, Obstacles and Pilot Projects", General Meeting of IEEE Power Engineering Society (PESGM).

- 2019 8. **Sossan, F.** (2019 November). Dispatching the operation of electrical distribution systems and providing multiple ancillary services to the power grid with grid-connected batteries. Seminar, Monash University, Australia.
- 2019 7. **Sossan, F.** (2019 May). Dispatch and clustering of ancillary services with distributed energy storage. Seminar, Tsinghua University (Prof. Z. Hu's lab), China.
- 2018 6. **Sossan, F.** (2018 August). Dispatch and Primary Frequency Control with Distributed Electrochemical Storage Systems: a System-wise Validation via Real-Time Simulation. Panel presentation, session "Real-Time Simulation and Testing of Multi-Domain Systems using Detailed Modeling and Experimental Validation", General Meeting of IEEE Power Engineering Society (PESGM).
- 2018 5. **Sossan, F.** (2018 June). Dispatch and clustering of ancillary services from distributed storage. Tutorial, opening session on "Modeling and applications of energy storage systems in power grids", PSCC.
- 2017 4. **Sossan, F.** (2017 December). Achieving the Dispatchability of Stochastic Power Flows by Distributed Control of Dispersed Energy Resources. Seminar, National Renewable Energy Laboratory (NREL), USA.
- 2017 3. **Sossan, F.**, & Paolone, M. (2017 July). Aggregation of Power Capabilities of Heterogeneous Resources for the Real-Time Optimal Control of Active Distribution Networks. Panel presentation, session "Modern Heuristic Optimization Techniques for Renewable Energy Sources Integration with Energy Storage Devices: Optimization Under Uncertainty", General Meeting of IEEE Power Engineering Society (PESGM).
- 2017 2. **Sossan, F.** (2017 March). Dispatching active distribution networks through electrochemical storage systems and demand side management. Seminar, University of Genova, Italy.
- 2016 1. **Sossan, F.** (2016 February). Evaluation of the impact of dispatched-by-design operation on power system reserve requirements. Invited Presentation, Future Electric Power Systems and the Energy Transition International conference, Champéry, Switzerland.

Honors, Distinctions and Recognition

- 2021 **CapEnergies accreditation.** Experimental infrastructure for integrated energy systems recognized by CapEnergies for research excellence and industrial impact.
- 2020 **Top 5% paper award.** Cassano, S., Nicolet, C., & **Sossan, F.**, "Reduction of Penstock Fatigue in a Medium-Head Hydropower Plant Providing Primary Frequency Control", UPEC 2020. [Publisher Link](#) — [Preprint](#).
- 2018 **Best paper award.** Valenciano López, A., Bozorg, M., **Sossan, F.**, & Paolone, M., "An Econometric Model of the Regulating Power Price for Interconnected Power Systems: The Case of the Nord Pool Market", EEM 2018. [Publisher Link](#) — [Preprint](#).
- 2013 **Best poster paper award.** **Sossan, F.**, Marinelli, M., Costanzo, G. T., & Bindner, H., "Indirect Control of DSRs for Regulating Power Provision and Solving Local Congestions", IYCE 2013. [Publisher Link](#) — [Preprint](#).

Media appearance

- 2024 [RSI](#) (Radiotelevisione Svizzera Italiana), while covering the academic offer of my university, featured a laboratory of my class.

2023 [PV magazine](#) covers the work of Biswarup Mukherjee and mine on deployment of EV charging stations.

Services to the profession

2019 – 2022 Associate Editor of Elsevier Sustainable Energy, Grids and Networks (SEGAN).
 2019 – present Member of the technical program committee of the Power Systems Computation Conference (PSCC).
 2018 – 2019 Member of the technical program committee of IEEE SmartGridComm and session chair.
 2014 – present Reviewer for leading journals and conferences in power systems, including IEEE Transactions on Sustainable Energy, IEEE Transactions on Smart Grid, IEEE Transactions on Industrial Informatics, PSCC, IEEE PowerTech, and IEEE ISGT.

Working groups and professional societies

2021 – present Member of CIGRE
 2021 – 2023 CIGRE working group C6.43, *Aggregation of Battery Energy Storage Systems*
 2020 – 2021 CIRED working group 2019-4, *Storage Technologies as an Opportunity for Distribution Systems*
 2018 – 2019 Member of the *ASI camera benchmark* campaign, in the context of the IEA PVPS Task 16 on *Solar Resource for High-penetration and Large-scale Applications*.
 2012 – 2025 Member of IEEE, and IEEE Power Engineering Society

Other services

2026 Ph.D. thesis jury member, DTU Technical University of Denmark (A. Malkova)
 2024 Ph.D. thesis jury member, KTH Royal Institute of Technology (W. Naim)
 2023 Ph.D. thesis jury member, EPFL (S. Fahmy)
 2022 Ph.D. thesis jury member, University of Genova (G. Mosaico)
 2022 Ph.D. thesis jury member, University of Genova (L. Briano)
 2020 Ph.D. thesis jury member, University of Genova (P. Pongiglione)
 2019 – present External member of the Ph.D. school board. Department of Power System Engineering and Transportation Systems, University of Genova.
 2019 – present Evaluator of research projects for several research funding bodies.

List of publications

(Reverse chronological order and reverse numbering)

Book chapters and Technical Brochures

2021 **Sossan, F.**, & Alvarado, F. (2022) Battery energy storage systems for applications in distribution grids, in book *Planning and Operation of Active Distribution Networks*, Springer International Publishing.

Peer-reviewed publications in scientific journals

2026 38. Id Omar, N. E., Lê-Agopyan, A., & **Sossan, F.** (2026). Control and scheduling of behind-the-meter battery energy storage systems for stacked grid and building services. *Smart Energy*, 100249. [Link to open repository](#). [Link to publisher](#)

- 2025 37. Cassano, S., & **Sossan, F.** (2025). Scheduling power-intensive operations of Battery Energy Storage Systems and application to hybrid hydropower plants. *Applied Energy*, 386, 125559. [Link to open repository](#). [Link to publisher](#)
- 2024 36. Cassano, S., & **Sossan, F.** (2024). Scheduling Power-Intensive Operations of Battery Energy Storage Systems and Application to Hybrid Hydropower Plants. Revision 0 is currently under review in *Applied Energy*. [Link to publisher](#)
- 2024 35. Grammatikos, P., Le Boudec, J. Y., Paolone, M., & **Sossan, F.** (2024). Computation of ultra-short-term prediction intervals of the power prosumption in active distribution networks. *Electric Power Systems Research*, 235, 110780. [Link to publisher](#)
- 2023 34. Gupta, R., & **Sossan, F.** (2023). Optimal Sizing and Siting of Energy Storage Systems Considering Curtailable Photovoltaic Generation in Power Distribution Networks. *Applied Energy*. [PDF from my website](#). [Link to publisher](#)
- 2023 33. Mukherjee, B., & **Sossan, F.** (2023). Optimized Planning of Chargers for Electric Vehicles in Distribution Grids Including PV Self-consumption and Cooperative Vehicle Owners. *IET Energy Conversion and Economics*. [PDF from my website](#). [Link to publisher](#)
- 2022 32. Ledur, S., Molinier, R., **Sossan, F.**, Alais, J. C., Faris, M. D. E. A., & Kariniotakis, G. (2022). Identification and quantification of the flexibility potential of a complex industrial process for ancillary services provision. *Electric Power Systems Research*, 212, 108396. [Link to open repository](#). [Link to publisher](#)
- 2022 31. Mukherjee, B., & **Sossan, F.** (2022). Optimal planning of single-port and multi-port charging stations for electric vehicles in medium voltage distribution networks. *IEEE Transactions on Smart Grid*. [PDF from my website](#). [Link to publisher](#)
- 2022 30. Cassano, S., & **Sossan, F.** (2022). Model Predictive Control for a Medium-head Hydropower Plant Hybridized with Battery Energy Storage to Reduce Penstock Fatigue. *Electric Power Systems Research*, 213, 108545. [PDF from my website](#). [Link to open repository](#)
- 2022 29. Gupta, R., **Sossan, F.**, & Paolone, M. (2022). Model-less Robust Voltage Control in Active Distribution Networks using Sensitivity Coefficients Estimated from Measurements. *Electric Power Systems Research/Power Systems Computation Conference (Accepted)*. [Link to open repository](#)
- 2022 28. Cassano, S., & **Sossan, F.** (2022). Stress-informed Control of Medium- and High-head Hydropower Plants to Reduce Penstock Fatigue. *Sustainable Energy, Grids and Networks*, 31, 100688. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2021 27. Gupta, R., **Sossan, F.**, & Paolone, M. (2021). Countrywide PV hosting capacity and energy storage requirements for distribution networks: The case of Switzerland. *Applied Energy*, Volume 281, 116010. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2021 26. Massucco, S., Paolone, M., Pongiglione, P., Silvestro, F., & **Sossan, F.** (2021). Siting and sizing of energy storage systems: Towards a unified approach for transmission and distribution system operators for reserve provision and grid support. *Electric Power Systems Research*, 190. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2021 25. Zecchino, A., Yuan, Z., **Sossan, F.**, Cherkaoui, R., & Paolone, M. (2021). Optimal Provision of Concurrent Primary Frequency and Local Voltage Control from a BESS Considering

- Variable Capability Curves: Modelling and Experimental Assessment. Electric Power Systems Research, 190. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2021 24. Zuo, Y., Yuan, Z., **Sossan, F.**, Zecchino, A., Cherkaoui, R., & Paolone, M. (2021). Performance assessment of grid-forming and grid-following converter-interfaced battery energy storage systems on frequency regulation in low-inertia power grids. Sustainable Energy, Grids and Networks. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2021 23. Gupta, R., **Sossan, F.**, Le Boudec, J. Y., & Paolone, M. (2021). Compound Admittance Matrix Estimation of Three-Phase Untransposed Power Distribution Grids Using Synchrophasor Measurements. IEEE Transactions on Instrumentation and Measurement, 70. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2020 22. Gupta, R., **Sossan, F.**, & Paolone, M. (2020). Grid-aware Distributed Model Predictive Control of Heterogeneous Resources in a Distribution Network: Theory and Experimental Validation. IEEE Transactions on Energy Conversion. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2020 21. Nespoli, L., Medici, V., Kristijan, L., & **Sossan, F.** (2020). Hierarchical demand forecasting benchmark for the distribution grid. Electric Power Systems Research, 189, 106755. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2020 20. Stai, E., **Sossan, F.**, Namor, E., Le Boudec, J.Y., & Paolone, M. (2020). A receding horizon control approach for re-dispatching stochastic heterogeneous resources accounting for grid and battery losses. Electric Power Systems Research, 185. [Link to publisher](#)
- 2020 19. Zuo, Y., **Sossan, F.**, & Paolone, M. (2020). Effect of voltage source converters with electrochemical storage systems on dynamics of reduced-inertia bulk power grids. Electric Power Systems Research, 189. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2019 18. Kalantar-Neyestanaki, M., **Sossan, F.**, Bozorg, M., & Cherkaoui, R. (2019). Characterizing the reserve provision capability area of active distribution networks: a linear robust optimization method. IEEE Transactions on Smart Grid, 11(3): 2464-2475. [Link to publisher](#)
- 2019 17. **Sossan, F.**, Scolari, E., Gupta, R., & Paolone, M. (2019). Solar irradiance estimations for modeling the variability of photovoltaic generation and assessing violations of grid constraints: A comparison between satellite and pyranometers measurements with load flow simulations. Journal of Renewable and Sustainable Energy of American Institute of Physics, 11(5). [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2018 16. Bozorg, M., **Sossan, F.**, Le Boudec, J.Y., & Paolone, M. (2018). Influencing the bulk power system reserve by dispatching power distribution networks using local energy storage. Electric Power Systems Research, 163:270 – 279. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2018 15. Fabietti, L., Gorecki, T. T., Namor, E., **Sossan, F.**, Paolone, M., & Jones, C. N. (2018). Enhancing the dispatchability of distribution networks through utility-scale batteries and flexible demand. Energy and Buildings, 172:125 – 138. [Link to open repository](#). [Link to publisher](#)
- 2018 14. Gao, X., **Sossan, F.**, Christakou, K., Paolone, M., & Liserre, M. (2018). Concurrent voltage control and dispatch of active distribution networks by means of smart transformer and storage. IEEE Transactions on Industrial Electronics, 65(8):6657–6666. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)

- 2018 13. Mahmood, F., Vanfretti, L., Pignati, M., **Sossan, F.**, & Paolone, M. (2018). Experimental validation of a steady state model synthesis method for a three-phase unbalanced active distribution network feeder. *IEEE Access*, 6:4042–4053. [Link to publisher](#)
- 2018 12. Namor, E., **Sossan, F.**, Cherkaoui, R., & Paolone, M. (2018). Control of battery storage systems for the simultaneous provision of multiple services. *IEEE Transactions on Smart Grid*, 10(3):2799–2808. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2018 11. Scolari, E., Reyes, L., **Sossan, F.**, & Paolone, M. (2018). A comprehensive assessment of the short-term uncertainty of grid-connected pv systems. *IEEE Transactions on Sustainable Energy*, 9(3):1458–1467. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2018 10. Scolari, E., **Sossan, F.**, Haure-Touze, M., & Paolone, M. (2018). Local estimation of the global horizontal irradiance using an all-sky camera. *Solar Energy*, 173:1225 – 1235. [Link to publisher](#)
- 2018 9. **Sossan, F.**, Nespoli, L., Medici, V., & Paolone, M. (2018). Unsupervised disaggregation of photovoltaic production from composite power flow measurements of heterogeneous prosumers. *IEEE Transactions on Industrial Informatics*, 14(9):3904–3913, 2018. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2018 8. Stai, E., Reyes-Chamorro, L., **Sossan, F.**, Le Boudec, J.Y., & Paolone, M. (2018). Dispatching stochastic heterogeneous resources accounting for grid and battery losses. *IEEE Transactions on Smart Grid*, 9(6):6522–6539. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2017 7. Scolari, E., **Sossan, F.**, & Paolone, M. (2017). Photovoltaic-model-based solar irradiance estimators: Performance comparison and application to maximum power forecasting. *IEEE Transactions on Sustainable Energy*, 9(1):35–44. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2017 6. **Sossan, F.** (2017). Equivalent electricity storage capacity of domestic thermostatically controlled loads. *Energy*, 122. [PDF from my website](#). [Link to publisher](#)
- 2016 5. Scolari, E., **Sossan, F.**, & Paolone, M. (2016). Irradiance prediction intervals for PV stochastic generation in microgrid applications. *Solar Energy*, 139. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2016 4. **Sossan, F.**, Lakshmanan, V., Costanzo, G. T., Marinelli, M., Douglass, P. J., & Bindner, H. (2016). Grey-box modelling of a household refrigeration unit using time series data in application to demand side management. *Sustainable Energy, Grids and Networks*, 5. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2016 3. **Sossan, F.**, Namor, E., Cherkaoui, R., & Paolone, M. (2016). Achieving the dispatchability of distribution feeders through prosumers data driven forecasting and model predictive control of electrochemical storage. *IEEE Transactions on Sustainable Energy*, 7(4):1762–1777. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2014 2. **Sossan, F.**, Bindner, H., Madsen, H., Torregrossa, D., Chamorro, L. R., & Paolone, M. (2014). A model predictive control strategy for the space heating of a smart building including cogeneration of a fuel cell-electrolyzer system. *International Journal of Electrical Power & Energy Systems*, 62:879 – 889. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)

- 2014 1. Marinelli, M., **Sossan, F.**, Costanzo, G. T., & Bindner, H. W. (2014). Testing of a predictive control strategy for balancing renewable sources in a microgrid. *IEEE Transactions on Sustainable Energy*, 5(4):1426–1433. [Link to open repository](#). [Link to publisher](#)

Peer-reviewed publications in conference proceedings

- 2025 40. Ali, A. M., Raeisi, Y., Grammatikos, P., Pavanello, D., Roduit, P., & **Sossan, F.** (2025). Large-Scale Processing and Validation of Grid Data for Assessing the Fair Spatial Distribution of PV Hosting Capacity. *arXiv preprint arXiv:2507.08684*.
- 2025 39. Grammatikos, P., Ali, A. M., & **Sossan, F.** (2025). Measurement-based line-impedance estimation in the absence of phasor measurement units. In 2025 IEEE Kiel PowerTech (pp. 1-6). IEEE. [Link to open repository](#). [Link to publisher](#)
- 2024 38. Hatziaargyriou, N., Joos, G., Skarvelis-Kazakos, S., Susanto, J., Offergeld, T., Kenarangui, Y., Verma, S., Ndiaye, I., Oulis-Rousis, A., Schwaegerl, C., Broderick, S., Reilly, Buchholz, B., Yamashita, K., Saridaki, G., **Sossan, F.**, Valliou, M., Blavette, A., Evans, H., Fan, L., & Wang, H. (2024). Aggregation of Battery Energy Storage and Distributed Energy Resources. *Cigre Technical Brochures*. [Link to publisher](#)
- 2023 37. Camal, S., van Der Meer, D., **Sossan, F.**, & Kariniotakis, G. (2023). Hierarchical Forecasting for the Management of Distribution Grids. In 27th International Conference on Electricity Distribution (CIRED 2023) (pp. 2233-2237). [Link to open repository](#)
- 2021 36. Mukherjee, B., Kariniotakis, G., & **Sossan, F.** (2021). Smart Charging, Vehicle-to-Grid, and Reactive Power Support from Electric Vehicles in Distribution Grids: A Performance Comparison. In 2021 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe). IEEE. [PDF from my website](#). [Link to open repository](#)
- 2021 35. Cassano, S., Landry, C., Nicolet, C., & **Sossan, F.** (2021). Performance Assessment of Linear Models of Hydropower Plants. In 2021 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe). IEEE. [PDF from my website](#). [Link to open repository](#)
- 2021 34. Gupta, R., Sovljanski, V., **Sossan, F.**, & Paolone, M. (2021). Performance Comparison of Alternating Direction Optimization Methods for Linear-OPF based Real-time Predictive Control. In 2021 IEEE Madrid PowerTech. IEEE. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2020 33. Cassano, S., Nicolet, C., & **Sossan, F.** (2020). Reduction of Penstock Fatigue in a Medium-Head Hydropower Plant Providing Primary Frequency Control. In 2020 55th International Universities Power Engineering Conference (UPEC). IEEE. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2020 32. **Sossan, F.**, Mukherjee, B., & Hu, Z. (2020). Impact of the Charging Demand of Electric Vehicles on Distribution Grids: a Comparison Between Autonomous and Non-Autonomous Driving. In 15th International Conference on Ecological Vehicles and Renewable Energies (EVER). IEEE. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
- 2019 31. Gupta, R., **Sossan, F.**, & Paolone, M. (2019). Performance assessment of linearized OPF-based distributed real-time predictive control. In 2019 IEEE Manchester PowerTech. IEEE. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)

- 2019 30. Kalantar-Neyestanaki, M., Bozorg, M., **Sossan, F.**, & Cherkaoui, R. (2019). Allocation of active power reserve from active distribution networks using a cost-benefit approach: Application to Swissgrid network. In 2019 IEEE Manchester PowerTech. IEEE. [Link to publisher](#)
- 2018 29. Gupta, R., **Sossan, F.**, Scolari, E., Namor, E., Fabietti, L., Jones, C. N., & Paolone, M. (2018). An ADMM-based coordination and control strategy for PV and storage to dispatch stochastic prosumers: Theory and experimental validation. In 2018 Power Systems Computation Conference (PSCC). [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
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- 2018 27. Namor, E., **Sossan, F.**, Scolari, E., Cherkaoui, R., & Paolone, M. (2018). Experimental assessment of the prediction performance of dynamic equivalent circuit models of grid-connected battery energy storage systems. In 2018 IEEE International Conference on Innovative Smart Grid Technologies (ISGT). IEEE. [PDF from my website](#). [Link to open repository](#). [Link to publisher](#)
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